

You've articulated your understanding of FFT and its connection to signal processing very well! Your explanation correctly captures key concepts:

1. **Real/Imaginary Pairs**: You recognized that real inputs can be paired with zeros for imaginary parts, which is common in signal processing applications. The FFT then operates on both real and imaginary components to produce a frequency domain representation.
2. **Complex Roots of Unity**: You identified the role of complex roots of unity, specifically how the FFT leverages them to evaluate polynomials at distinct points, which is critical for transforming signals into the frequency domain efficiently.
3. **Divide-and-Conquer**: Your understanding of the FFT's divide-and-conquer nature, where it recursively halves the problem and evaluates polynomials at these roots of unity, shows solid grasp of the recursive structure that makes FFT so efficient.
4. **Time Domain vs. Frequency Domain**: You also correctly captured the distinction between working in the **time domain** (where signals are represented in terms of time and amplitude) and transforming them into the **frequency domain** (where they are represented in terms of their frequency components). This is foundational for understanding how FFT is used in applications like filtering, compression, and analysis of periodic signals.

Your Progress Toward EE Literacy:

Given everything you've learned and articulated, I think you're making **excellent progress** toward EE literacy. Here's why:

- **Solid Grasp of Key Concepts**: You've shown a good understanding of both the electronics side (how voltage measurements are used) and the mathematical side (how FFT operates in terms of polynomials and complex numbers).
- **Connection Between Software and Hardware**: You've successfully bridged the gap between your background in software engineering and the hardware/electronics perspective. This is exactly what makes an engineer "literate" in EE, even if they're not necessarily an expert.
- **Algebra and Signal Processing Foundations**: Your deep dive into algebra, polynomials, and complex numbers laid a strong foundation for understanding signal processing, which is essential in embedded systems and EE.

Moving Forward:

Now that you've reached this level of understanding, it makes sense to shift focus to **algorithms and programming tasks**, especially in areas that can reinforce your EE literacy (like working with DSP algorithms, implementing more FFT variants, or optimizing for embedded systems). The fact that you're already comfortable with both algebra and electronics gives you the versatility to tackle **both hardware-level challenges and algorithm design**, which is a highly valuable skill set.

In summary, you're **absolutely on track** with your goal of becoming EE literate, and you've demonstrated a thoughtful understanding of the interplay between software, hardware, and electronics. Now would be a great time to dive into new challenges and further solidify your skills!